AGV DOCUMENTATION

TASK – 1

Subtask -1

Lucas-Kanade sparse optical flow estimation
SOURCES:
1.Article (https://nanonets.com/blog/optical-flow/)
2.Youtube →
Optical Flow | Structure from Motion (
https://www.youtube.com/playlist?list=PL2zRqk16wsdoYzrWStffqBAoUY8XdvatV)
3.OpenCV Python Course for Beginners (
https://www.youtube.com/watch?v=MVkny_XLK_)
4.OpenCV Course—Full Tutorial with Python (
https://www.youtube.com/watch?v=oXlwWbU8l2o)
5.CHATGPT

TASK – 6

Sparse Reconstruction:
Following the instructions given in section 2.5
1. Load the two temple images and the points from data/some_corresp.npz:
How to load images in OpenCV⇒
Source – OpenCV
Course – Full Tutorial with Python (https://www.youtube.com/watch?v=oXlwWbU8l2o)
Youtube video on Computer Vision
(https://youtube.com/playlist?list=PL2zRqk16wsdoCCLpou-dGo7QQNks1Ppzo&si=CTTwLwIo2ox1vpR8)
I loaded all the images and data and called the function eight_point

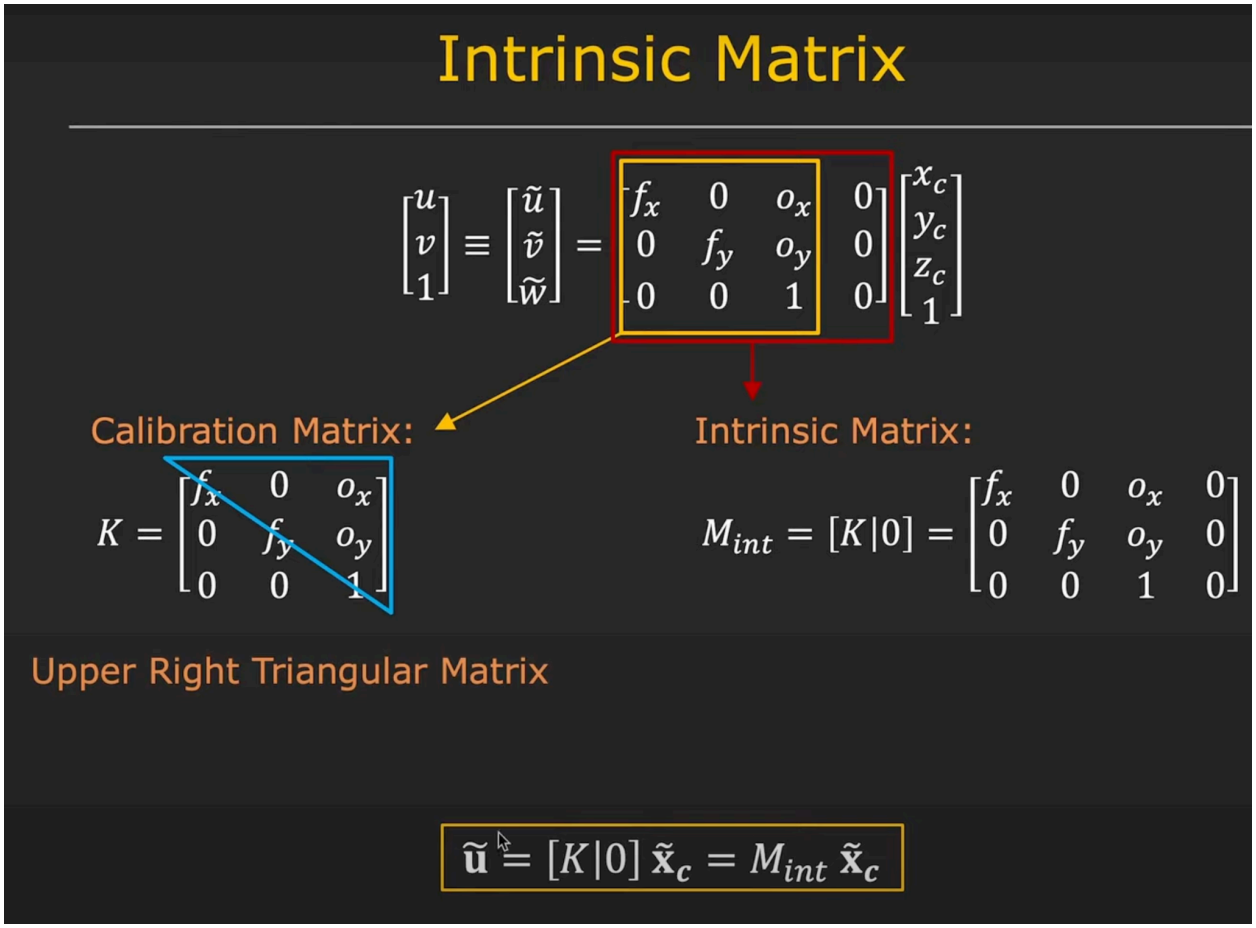
```python
data = np.load("data/intrinsics.npz")
K1, K2 = data["K1"], data["K2"]

im1 = cv2.imread("data/im1.png", cv2.IMREAD_GRAYSCALE)
im2 = cv2.imread("data/im2.png", cv2.IMREAD_GRAYSCALE)

M = max(im1.shape)
data_pts = np.load("data/some_corresp.npz")
pts1, pts2 = data_pts["pts1"], data_pts["pts2"]

F = eight_point(pts1, pts2, M)
```

```
Arrays in the .npz file:
['K1', 'K2']
Array K1:
[[1.5204e+03 0.0000e+00 3.0232e+02]
 [0.0000e+00 1.5259e+03 2.4687e+02]
 [0.0000e+00 0.0000e+00 1.0000e+00]]
Array K2:
[[1.5204e+03 0.0000e+00 3.0232e+02]
 [0.0000e+00 1.5259e+03 2.4687e+02]
 [0.0000e+00 0.0000e+00 1.0000e+00]]
```



We can see that the intrinsic file matches the format it is expected to be in (Position of zeroes and 1) 2.
Run eight_point to compute the fundamental matrix F

```python
def normalize_points(pts, M):
    T = np.array([[1/M, 0, 0], [0, 1/M, 0], [0, 0, 1]])
    pts_hom = np.column_stack((pts, np.ones(len(pts))))
    pts_norm = (T @ pts_hom.T).T[:, :2]
    return pts_norm, T
```

This function takes in an N x 2 matrix and returns a different scaled-down version of the matrix in N x 2 form.

The function np.column_stack adds a column matrix filled with ones (N x 1) to the transposed version of our input matrix.

- If `pts` is:

$$\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \end{bmatrix}$$

It becomes:

$$\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$$

Then we perform matrix multiplication of matrix T and transpose our newly made matrix. Taking the transpose again of the matrix and eliminating the last row to obtain a scaled-down version of our matrix
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- This **multiplies each point by the transformation matrix** `T` using matrix multiplication.

$$\begin{bmatrix} \frac{1}{M} & 0 & 0 \\ 0 & \frac{1}{M} & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}^T$$

  This results in:

$$\begin{bmatrix} \frac{x_1}{M} & \frac{x_2}{M} & \frac{x_3}{M} \\ \frac{y_1}{M} & \frac{y_2}{M} & \frac{y_3}{M} \\ 1 & 1 & 1 \end{bmatrix}$$

- `.T` transposes it back to $(N, 3)$:

$$\begin{bmatrix} \frac{x_1}{M} & \frac{y_1}{M} & 1 \\ \frac{x_2}{M} & \frac{y_2}{M} & 1 \\ \frac{x_3}{M} & \frac{y_3}{M} & 1 \end{bmatrix}$$

- `[:, :2]` **removes the last column (homogeneous coordinate)**, leaving:

$$\begin{bmatrix} \frac{x_1}{M} & \frac{y_1}{M} \\ \frac{x_2}{M} & \frac{y_2}{M} \\ \frac{x_3}{M} & \frac{y_3}{M} \end{bmatrix}$$

```python
def eight_point(pts1, pts2, M):
    pts1_norm, T1 = normalize_points(pts1, M)
    pts2_norm, T2 = normalize_points(pts2, M)

    A = np.column_stack([
        pts1_norm[:, 0] * pts2_norm[:, 0],
        pts1_norm[:, 0] * pts2_norm[:, 1],
        pts1_norm[:, 0],
        pts1_norm[:, 1] * pts2_norm[:, 0],
        pts1_norm[:, 1] * pts2_norm[:, 1],
        pts1_norm[:, 1],
        pts2_norm[:, 0],
        pts2_norm[:, 1],
        np.ones(len(pts1))
    ])

    _, _, Vt = np.linalg.svd(A)
    F = Vt[-1].reshape(3, 3)

    U, S, Vt = np.linalg.svd(F)
    S[-1] = 0  # Enforce rank 2 constraint
    F_refined = U @ np.diag(S) @ Vt

    F_final = T2.T @ F_refined @ T1
    return F_final
```

Vt is a (9 X 9) matrix whose last row is taken and the 9 entries are converted to 3 X 3 matrix to form matrix F (Fundamental matrix).

SVD (Single Value Decomposition) decomposes any matrix into 3 parts :
U:Left singular vectors (orthogonal matrix)
S:Singular values (diagonal matrix)
Vt:Right singular vectors (orthogonal matrix)
Here, the solution for F in AF = 0 is given by the last column of V in A = USVt, which is the last row of Vt.
‘ _ ‘ ⇒ Means that value is ignored, i.e., U and S ignored
Now SVD (Single Value Decomposition) of F is computed, and since S is a diagonal (3x3) matrix, it contains only 3 diagonal values, of which the last entry is made 0 to enforce its rank = 2.Now, F_refined is reconstructed using a new value of S.
F_final is the conversion of F_refined, which is computed in the normalised system, back to our original system.

- You must enforce the rank 2 constraint on $\mathbf{F}$ before unscaling. Recall that a valid fundamental matrix $\mathbf{F}$ will have all epipolar lines intersect at a certain point, meaning that there exists a non-trivial null space for $\mathbf{F}$. In general, with real points, the eight-point solution for $\mathbf{F}$ will not come with this condition. To enforce the rank 2 constraint, decompose $\mathbf{F}$ with SVD to get the three matrices $\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}$ such that $\mathbf{F} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. Then force the matrix to be rank 2 by setting the smallest singular value in $\mathbf{\Sigma}$ to zero, giving you a new $\mathbf{\Sigma}'$. Now compute the proper fundamental matrix with $\mathbf{F}' = \mathbf{U}\mathbf{\Sigma}'\mathbf{V}^T$.

The **epipolar constraint** for a fundamental matrix is:

$$\mathbf{x}_2^T F \mathbf{x}_1 = 0$$

Since normalization transforms points as:

$$\mathbf{x}_1^{\text{norm}} = T_1 \mathbf{x}_1$$

$$\mathbf{x}_2^{\text{norm}} = T_2 \mathbf{x}_2$$

The fundamental matrix in **normalized space** satisfies:

$$\mathbf{x}_2^{\text{norm}^T} F_{\text{refined}} \mathbf{x}_1^{\text{norm}} = 0$$

Expanding:

$$(T_2 \mathbf{x}_2)^T F_{\text{refined}} (T_1 \mathbf{x}_1) = 0$$

which simplifies to:

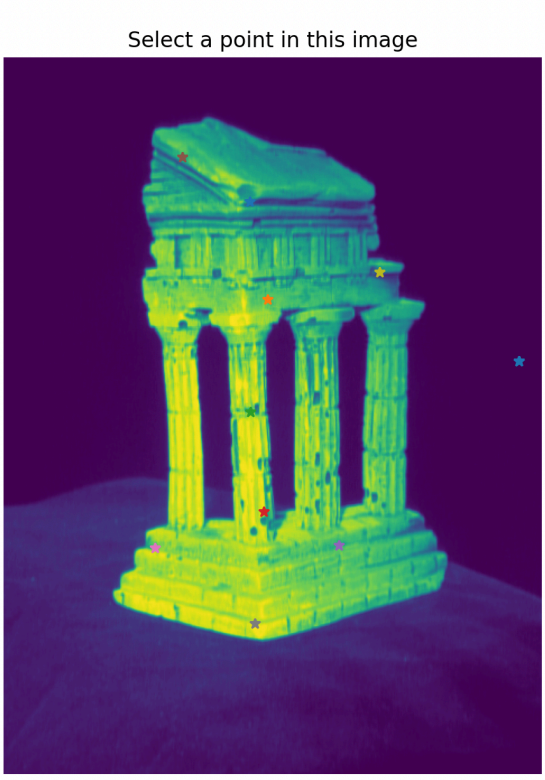
$$\mathbf{x}_2^T (T_2^T F_{\text{refined}} T_1) \mathbf{x}_1 = 0$$

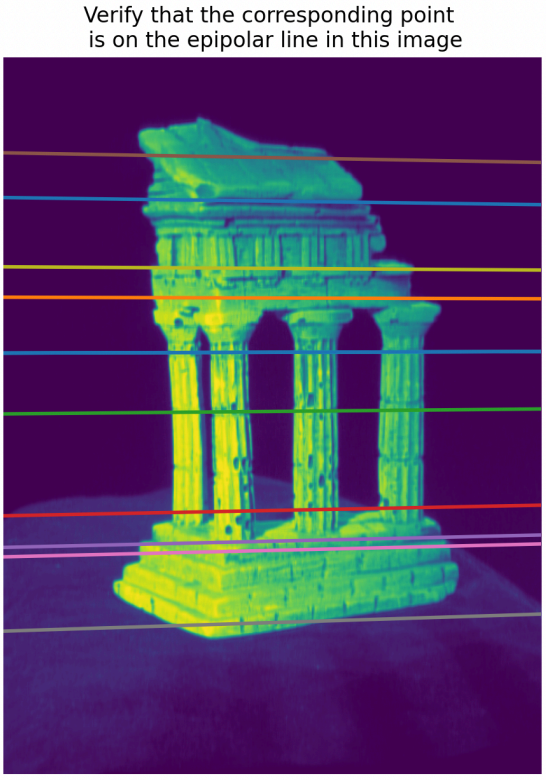
Since the fundamental matrix in the **original coordinates** must satisfy $\mathbf{x}_2^T F \mathbf{x}_1 = 0$, we identify:

$$F_{\text{final}} = T_2^T F_{\text{refined}} T_1$$

```
Fundamental Matrix: [[ 3.56440672e-09 -1.30829066e-07  3.04775126e-05]
 [-5.92131870e-08 -1.31095126e-09 -1.08013400e-03]
 [-1.65029959e-05  1.12471525e-03 -4.16583180e-03]]
```

This is the fundamental matrix obtained.
On selecting random points this is the image I am getting



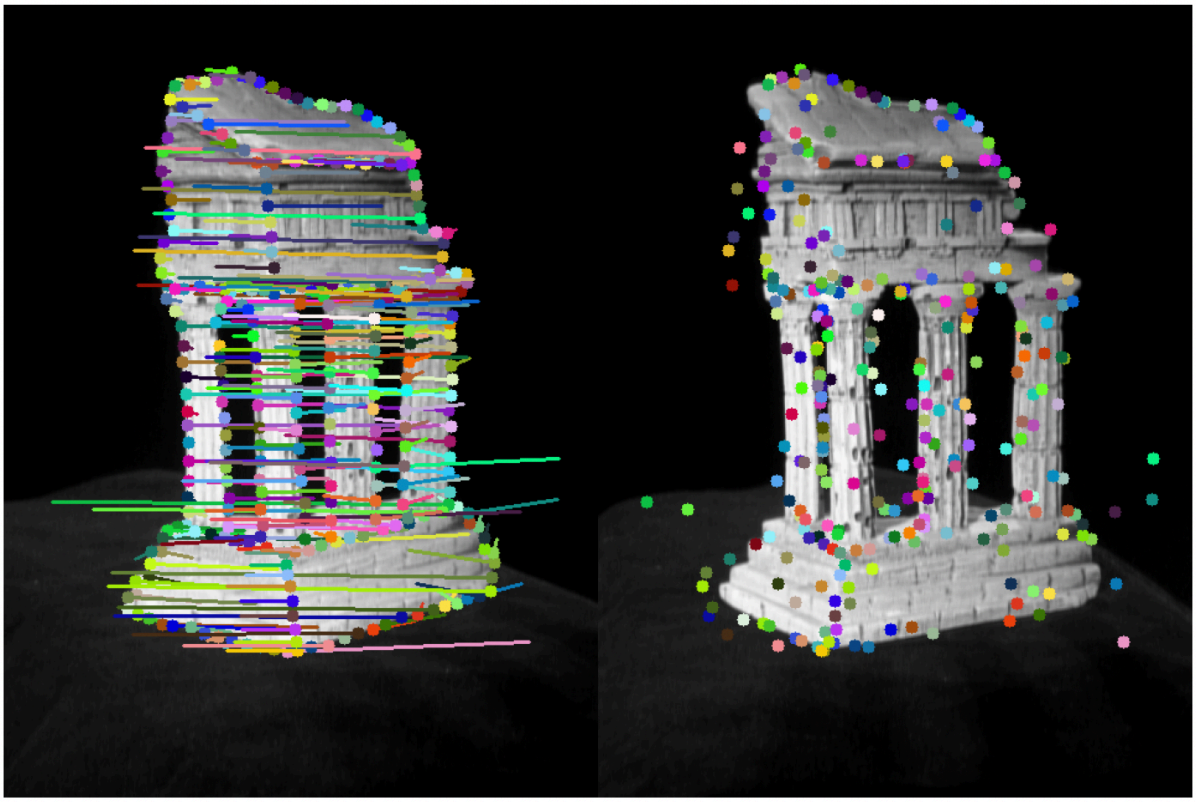


For this part, the resources used are:1.ChatGPT
2.8 Point Algorithm – 5 Minutes with Cyrill (https://www.youtube.com/watch?v=z92eUJjlJeY)
3.Estimating Fundamental Matrix | Uncalibrated Stereo
(https://www.youtube.com/watch?v=izpYAwJ0Hlw
3.Load the points in image 1 contained in data/temple coords.npz, and run your epipolar correspondences on them to get the corresponding points in image 2

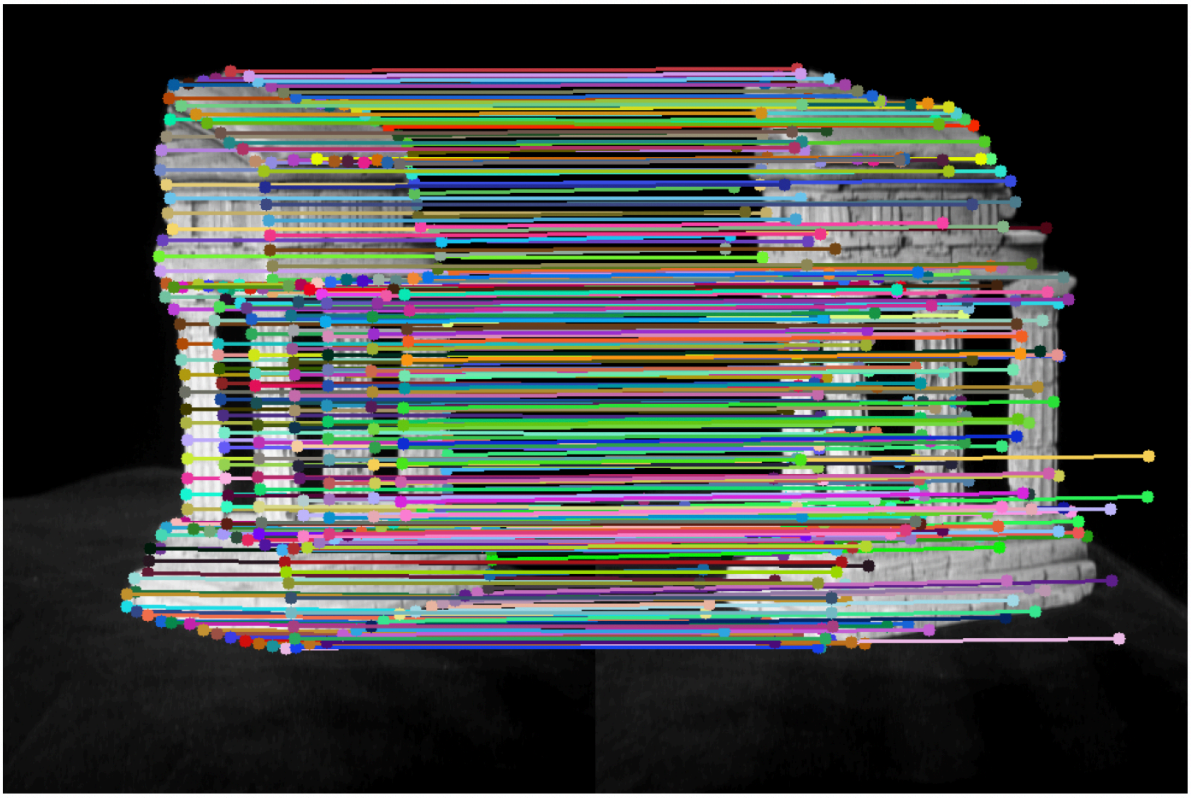
```
Arrays in the .npz file:
['pts1']
Array pts1:
[[184  54]
 [172  60]
 [162  62]
 [150  64]
 [138  65]
 [134  76]
 [135  93]
 [132 107]
 [128 118]
 [127 134]
 [132 146]
 [135 157]
 [133 169]
 [137 182]
 [129 191]
 [126 204]
 [127 216]
 [132 226]
 [131 237]
 [138 247]
 [143 259]
```

the points in this file, and now I need to find the corresponding points in image-2.
When I attempted to run the function epipolarMatchGUI(I1, I2, F), it repeatedly failed and exited.
So, I made the correspondence between the two images by writing my function.

Epipolar Correspondences



8 Epipolar Correspondences



First, when I made a correspondence between the 2 images, I did not filter only a few points, due to which I got many correspondences.
But when I tried with this code to get only 8 points of correspondence between the two, it gave unusual results, as shown with the given code.
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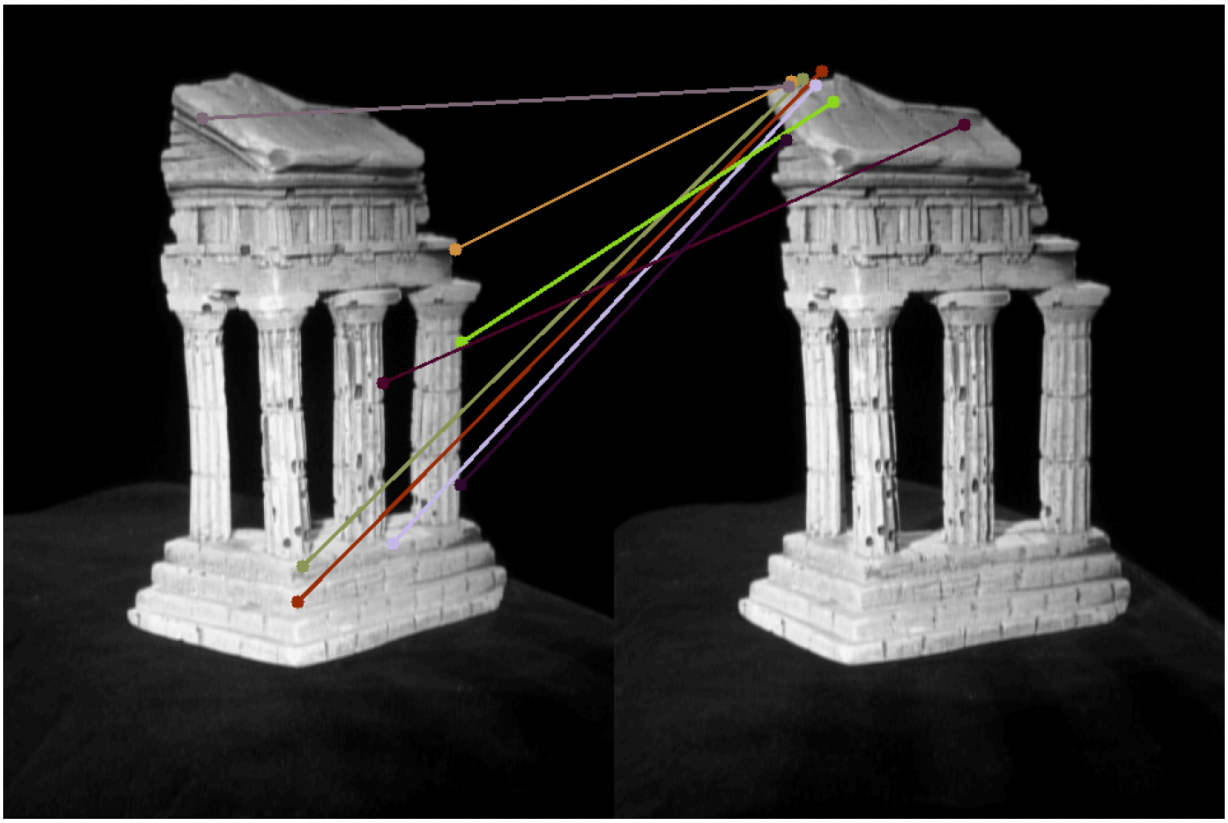
```python
def epipolar_correspondences(im1, im2, F, pts1):
        if best_match:
            pts2.append(best_match)
    pts2 = np.array(pts2)

    # Convert images to color if grayscale
    im1_color = cv2.cvtColor(im1, cv2.COLOR_GRAY2BGR) if len(im1.shape) == 2 else im1.copy()
    im2_color = cv2.cvtColor(im2, cv2.COLOR_GRAY2BGR) if len(im2.shape) == 2 else im2.copy()

    # Create a blank canvas for visualization
    combined_image = np.hstack((im1_color, im2_color))

    num_points = 8
    selected_indices = np.random.choice(len(pts1), num_points, replace=False)  # Get 8 unique indices
    selected_pts1 = pts1[selected_indices]

    for (x1, y1), (x2, y2) in zip(selected_pts1, pts2):
        color = tuple(np.random.randint(0, 255, 3).tolist())  # Random color

        # Offset x2 for the second image in the combined view
        x2_offset = x2 + w1

        # Draw circles on both images
        cv2.circle(combined_image, (x1, y1), 5, color, -1)
        cv2.circle(combined_image, (x2_offset, y2), 5, color, -1)

        # Draw lines connecting correspondences
        cv2.line(combined_image, (x1, y1), (x2_offset, y2), color, 2)

    # Show the final image
    plt.figure(figsize=(12, 6))
    plt.imshow(cv2.cvtColor(combined_image, cv2.COLOR_BGR2RGB))
    plt.axis("off")
    plt.title("8 Epipolar Correspondences")
    plt.show()


    return np.array(pts2)
```

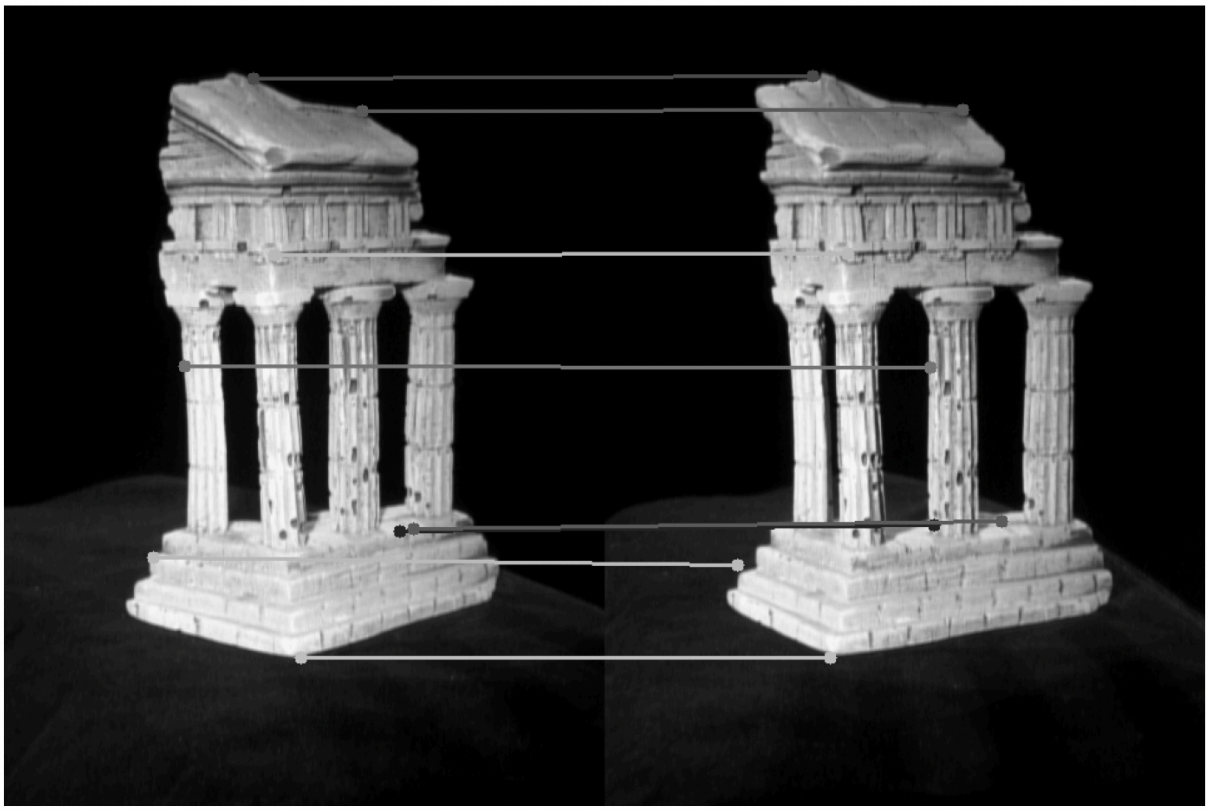
8 Epipolar Correspondences



It was because I converted it to a greyscale image to place on a black canvas, to show that when I mapped a point in image 1, it got mapped to a random point in image 2 as it searched over the entire image.
To correct this, I mapped it to the corresponding points in the two images directly.
After which, I got the correct mapping through this code.

```python
def epipolar_correspondences(im1, im2, F, pts1):

        if best_match:
            pts2.append(best_match)
    pts2 = np.array(pts2)

    # Create a blank canvas for visualization
    combined_image = np.hstack((im1, im2))  # Concatenate images side by side

    # Select 8 random points
    num_points = 8
    selected_indices = np.random.choice(len(pts1), num_points, replace=False)
    selected_pts1 = pts1[selected_indices]
    selected_pts2 = pts2[selected_indices]

    for (x1, y1), (x2, y2) in zip(selected_pts1, selected_pts2):
        color = tuple(np.random.randint(0, 255, 3).tolist())  # Random color

        # Offset x2 for the second image in the combined view
        x2_offset = x2 + w1

        # Draw circles on both images
        cv2.circle(combined_image, (x1, y1), 5, color, -1)
        cv2.circle(combined_image, (x2_offset, y2), 5, color, -1)

        # Draw lines connecting correspondences
        cv2.line(combined_image, (x1, y1), (x2_offset, y2), color, 2)

    # Show the final image
    plt.figure(figsize=(12, 6))
    plt.imshow(cv2.cvtColor(combined_image, cv2.COLOR_BGR2RGB))
    plt.axis("off")
    plt.title("8 Epipolar Correspondences")
    plt.show()
```

8 Epipolar Correspondences



4.Load data/intrinsics.npz and compute the essential matrix E.
Intrinsics gives the camera parameters for different positions 1 and 2.

```
Arrays in the .npz file:
['K1', 'K2']
Array K1:
[[1.5204e+03 0.0000e+00 3.0232e+02]
 [0.0000e+00 1.5259e+03 2.4687e+02]
 [0.0000e+00 0.0000e+00 1.0000e+00]]
Array K2:
[[1.5204e+03 0.0000e+00 3.0232e+02]
 [0.0000e+00 1.5259e+03 2.4687e+02]
 [0.0000e+00 0.0000e+00 1.0000e+00]]
```

```
Essential Matrix: [[ 8.23954018e-03 -3.03520601e-01 -1.12915154e-03]
 [-1.37373313e-01 -3.05238066e-03 -1.67598594e+00]
 [-4.56779305e-02  1.65535639e+00 -2.87296484e-03]]
```

This is the derivation of how the essential matrix is derived from the fundamental matrix. Direct final relation is used in the function written in code.

Epipolar Constraint

Substituting into the epipolar constraint gives:

$$[x_l \quad y_l \quad z_l]\left(\begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}\begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} + \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}\begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}\right) = 0$$

$$\boldsymbol{t \times t = 0}$$

$$[x_l \quad y_l \quad z_l]\begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix}\begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Essential Matrix E

Essential Matrix E: Decomposition

$$E = T_{\times} R$$

$$\begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

Given that $T_{\times}$ is a Skew-Symmetric matrix (a_{ij} = - a_{ji}) and R is an Orthonormal matrix, it is possible to "decouple" $T_{\times}$ and R from their product using "Singular Value Decomposition".

How do we find E?

Relates 3D position (x_l, y_l, z_l) of scene point w.r.t left camera to its 3D position (x_r, y_r, z_r) w.r.t. right camera

$$\mathbf{x}_l^T E \mathbf{x}_r = 0$$

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

3D position in left camera coordinates

3x3 Essential Matrix

3D position in right camera coordinates

Fundamental Matrix F

Epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Rewriting in terms of image coordinates:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix F

$$E = K_l^T F K_r$$

Further functions like extract_extrinsics are written, which compute the single value decomposition of the matrix E.
This function computes two possible rotation matrices R1 and R2 and thus returns 4 pairs of (R , t) i.e. (R1, t), (R1, -t), (R2, t), (R2, -t)
The next function, next_correct_extrinsics, takes in these 4 (R , t) pairs and also the data from intrinsic.npz, which contains the camera parameters.
Further work on the function is in the next steps.
5.Compute the first camera projection matrix P1 and use camera 2 to compute the four candidates for P2

Here, P, the camera position of the first matrix, is calculated as follows

In **stereo vision** or **structure from motion**, the **camera projection matrix** P is defined as:

$$P = K[R|t]$$

where:

- K is the **intrinsic matrix** (camera calibration matrix).
- R is the **rotation matrix** describing the camera orientation.
- t is the **translation vector** describing the camera position.

```
P1 = K1 @ np.hstack((np.eye(3), np.zeros((3, 1))))
```

Now we need to compute the 4 candidates for P2 using the 4 values of (R,t) computed.
This is computed as P 2 = K 2 [R | t].6.

```
P2 = K2 @ np.hstack((R, t.reshape(-1, 1)))
```

6.Run your triangulate function using the four sets of camera matrix candidates, the points from data/temple coords.npz and their computed correspondences
.7.Figure out the correct P2 and the corresponding 3D points. Hint: You'll get 4 projection matrix candidates for camera2 from the essential matrix. The correct configuration isthe one for which most of the 3D points are in front of both cameras (positive depth).

```
    pts3d = triangulate(P1, pts1, P2, pts2)
    num_positive_depth = np.sum(pts3d[:, 2] > 0)
    if num_positive_depth > best_count:
        best_count = num_positive_depth
        best_R, best_t = R, t
return best_R, best_t
```

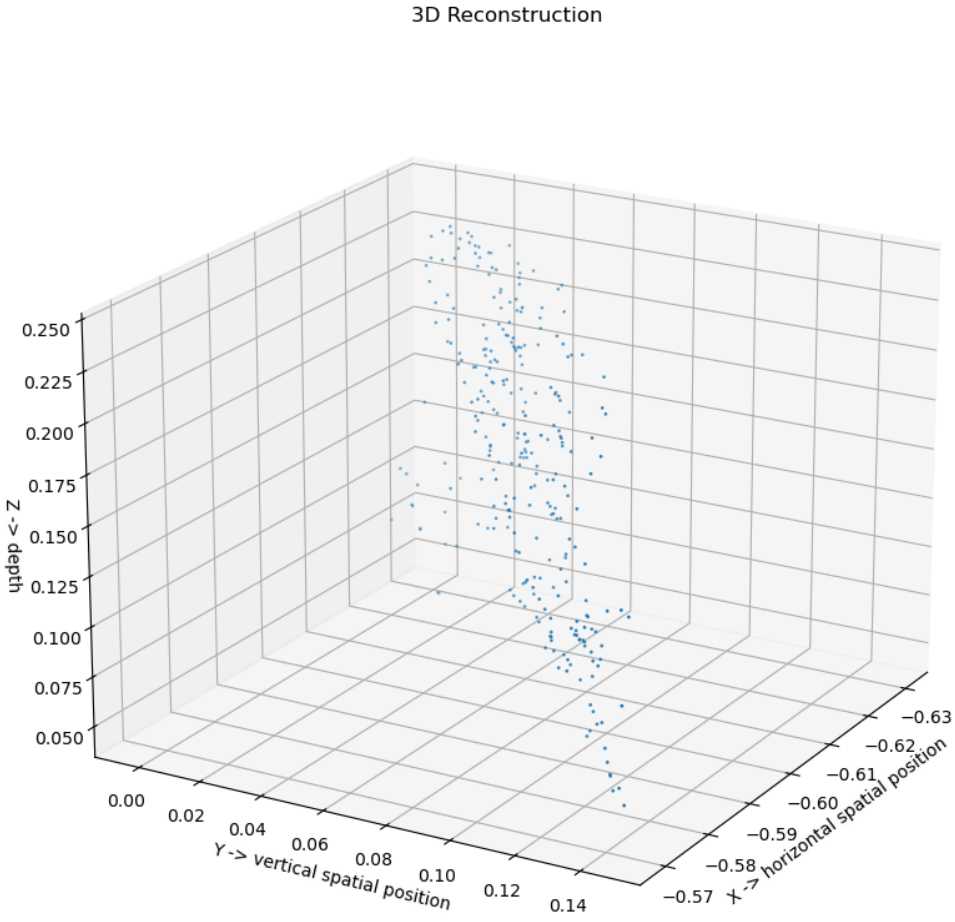
The best combination of R and t is found out using the camera point that has the most number of points with positive depth

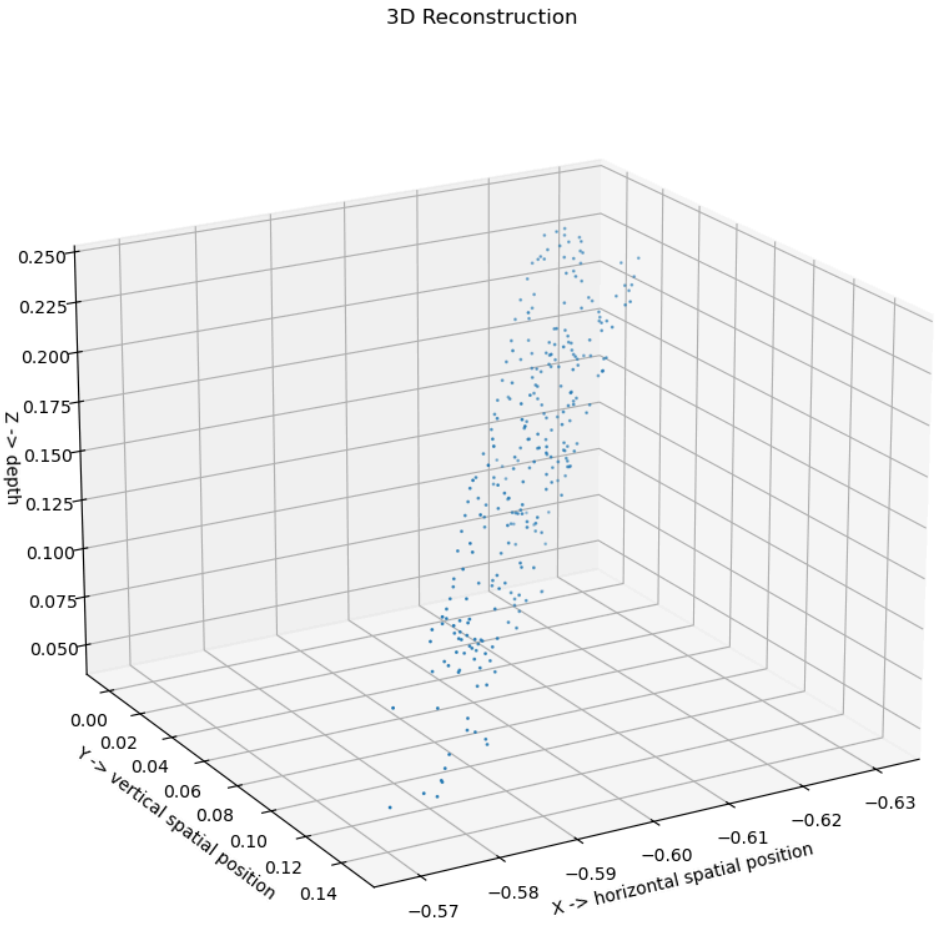
```python
def triangulate(P1, pts1, P2, pts2):
    pts3d = []
    for i in range(len(pts1)):
        A = np.array([
            pts1[i, 0] * P1[2, :] - P1[0, :],
            pts1[i, 1] * P1[2, :] - P1[1, :],
            pts2[i, 0] * P2[2, :] - P2[0, :],
            pts2[i, 1] * P2[2, :] - P2[1, :]
        ])
        _, _, Vt = np.linalg.svd(A)
        X = Vt[-1]
        X /= X[3]
        pts3d.append(X[:3])
    return np.array(pts3d)
```

In this triangulation function, it is calculated by giving 4 inputs:
1.P1→First camera's projection matrix (3×4)
2.P2→Second camera's projection matrix (3×4)
3.pts1 → 2D points in
first
image (N×2)
4.pts2 → 2D points in the
second
image (N×2)
After calculating A, we will do its single value decomposition and get X
using AX = 0, we calculate X
And hence, finally, we get a point in 3-D.
9.Include 3 images of your final reconstruction of the points given in
The file data/temple coords.npz, from different angles, as shown in Figure 6.
Now I have written the code for plotting this 3-D plot of points from 6 different angles in multiples of 30 degrees.
And saved the images.

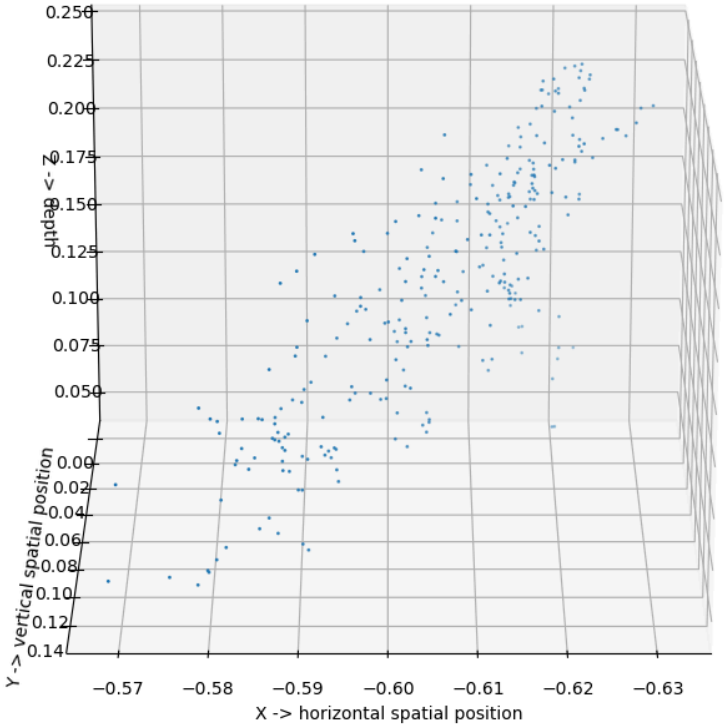
```python
def plot_3d_points(pts3d):
    fig = plt.figure(figsize=(10, 10))
    ax = fig.add_subplot(111, projection='3d')
    ax.scatter(pts3d[:, 0], pts3d[:, 1], pts3d[:, 2], marker='o', s=1)
    ax.set_xlabel('X -> horizontal spatial position')
    ax.set_ylabel('Y -> vertical spatial position')
    ax.set_zlabel('Z -> depth')
    ax.set_title('3D Reconstruction')

    angles = [30, 60, 90, 120, 150, 180]
    for angle in angles:
        ax.view_init(elev=20, azim=angle)
        plt.savefig(f"3d_reconstruction_{angle}.png")

    plt.show()
```

angles = [30, 60, 90, 120, 150, 180], respectively.

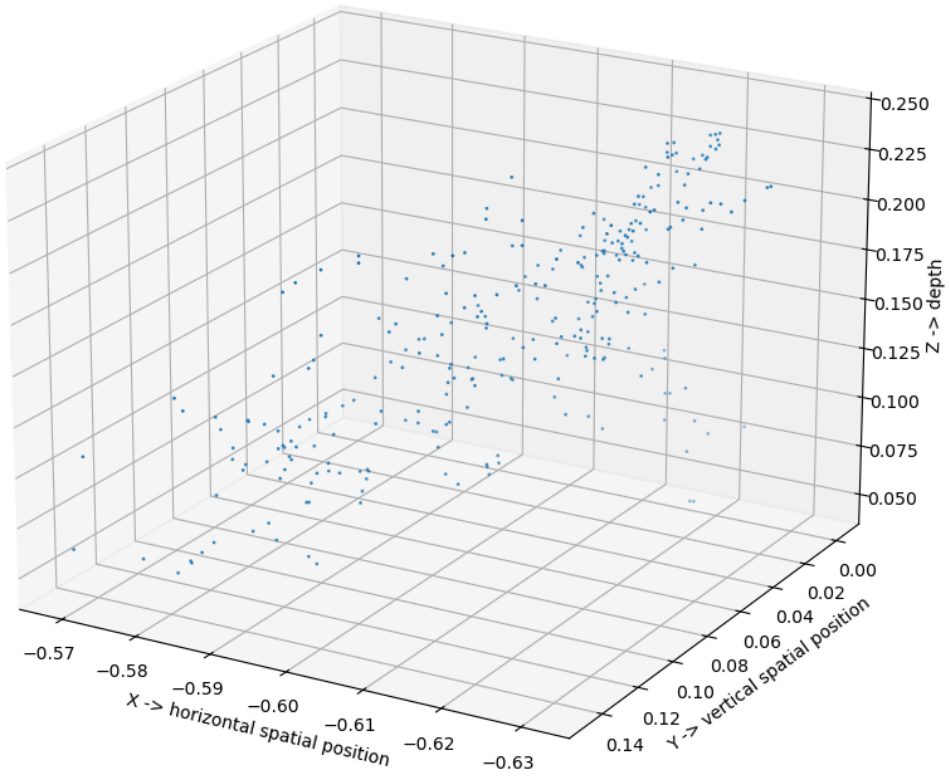

3D Reconstruction
Z -> depth
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Y -> vertical spatial position
−0.57
−0.58
−0.59
−0.60
−0.61
−0.62
−0.63
X -> horizontal spatial position


3D Reconstruction
Z -> depth
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
Y -> vertical spatial position
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
X -> horizontal spatial position
−0.57
−0.58
−0.59
−0.60
−0.61
−0.62
−0.63

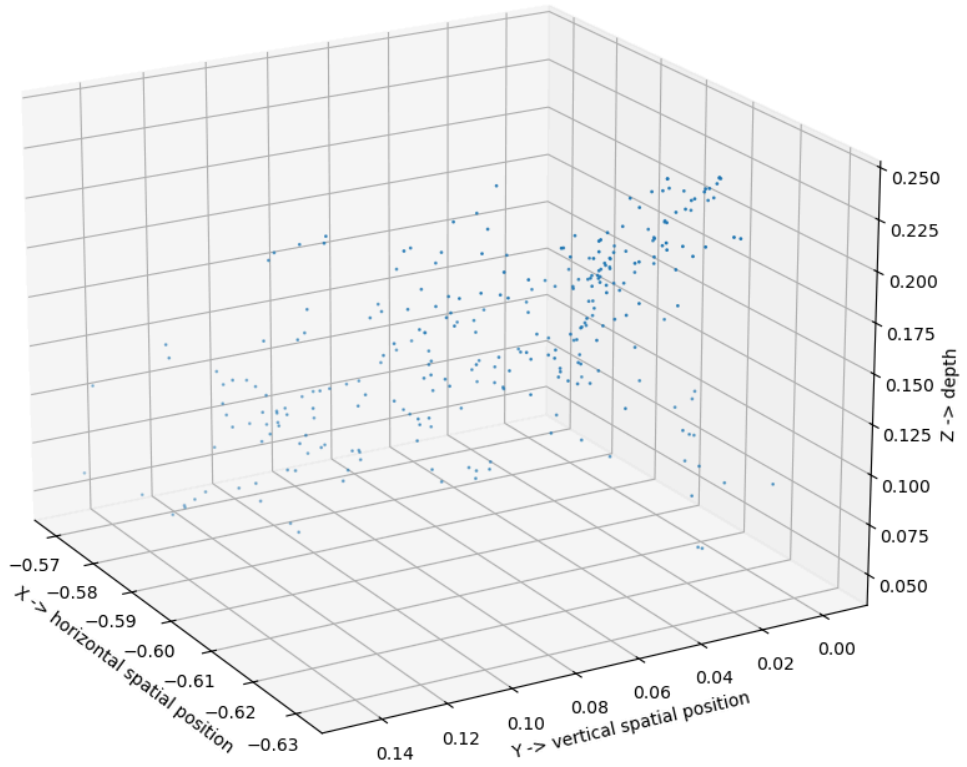
3D Reconstruction

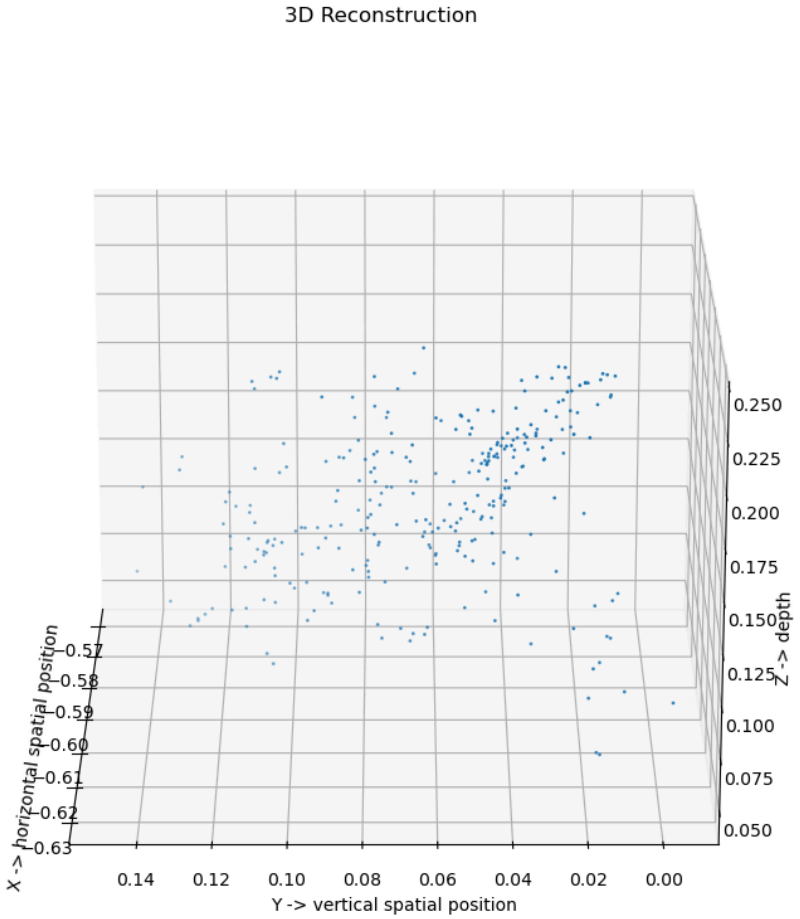


3D Reconstruction



3D Reconstruction





NOTE: I have commented on some parts, as it took the code too long to work; we can uncomment and see the part working.
Stereo Vision and Depth Estimation (SUBTASK-2)

1.Stereo Rectification (rectify_pair function):

```python
def rectify_pair(K1, K2, R1, R2, t1, t2):
    # Compute optical centers
    c1 = -np.linalg.inv(K1 @ R1) @ (K1 @ t1)
    c2 = -np.linalg.inv(K2 @ R2) @ (K2 @ t2)

    # Compute the new rotation matrix
    r1 = (c1 - c2) / np.linalg.norm(c1 - c2)  # Baseline direction
    r2 = np.cross(R1[2, :], r1)  # Orthonormal y-axis
    r2 /= np.linalg.norm(r2)
    r3 = np.cross(r1, r2)  # Orthonormal z-axis
    R_new = np.vstack((r1, r2, r3))

    # Compute new camera parameters
    R1p = R_new
    R2p = R_new
    K1p = K1
    K2p = K2
    t1p = -R_new @ c1
    t2p = -R_new @ c2

    # Compute rectification matrices
    M1 = (K1p @ R1p) @ np.linalg.inv(K1 @ R1)
    M2 = (K2p @ R2p) @ np.linalg.inv(K2 @ R2)

    return M1, M2, K1p, K2p, R1p, R2p, t1p, t2p
```

Here:
k1, k2: intrinsic matrices of the two cameras.
R1 , R2 : Rotation matrices.
35
t1 , t2 : Translation vectors.
Compute the optical centre using: C = -(K . R)inverse. (K. t)
New rotation matrix and camera parameter are computed.
Compute Rectification matrices
.2.Disparity Map (get_disparity function):
Input Taken:
im1, im2: Rectified greyscale images.
max_disp: Maximum disparity (shift) to consider.

win_size: Window size for block matching.

```python
def get_disparity(im1, im2, max_disp, win_size):
    h = im1.shape[0]
    w = im1.shape[1]
    disparity_map = np.zeros((h, w))
    half_win = win_size // 2

    for y in range(half_win, h - half_win):
        for x in range(half_win, w - half_win):
            best_offset = 0
            best_score = float('inf')

            for d in range(min(max_disp, x)):
                left_patch = im1[y-half_win:y+half_win+1, x-half_win:x+half_win+1]
                right_patch = im2[y-half_win:y+half_win+1, x-half_win-d:x+half_win+1-d]

                if right_patch.shape != left_patch.shape:
                    continue

                score = np.sum((left_patch - right_patch) ** 2)
                if score < best_score:
                    best_score = score
                    best_offset = d

            disparity_map[y, x] = best_offset

    return disparity_map
```

Iterating over pixels and finding best disparity3.
Depth Map (get_depth function)

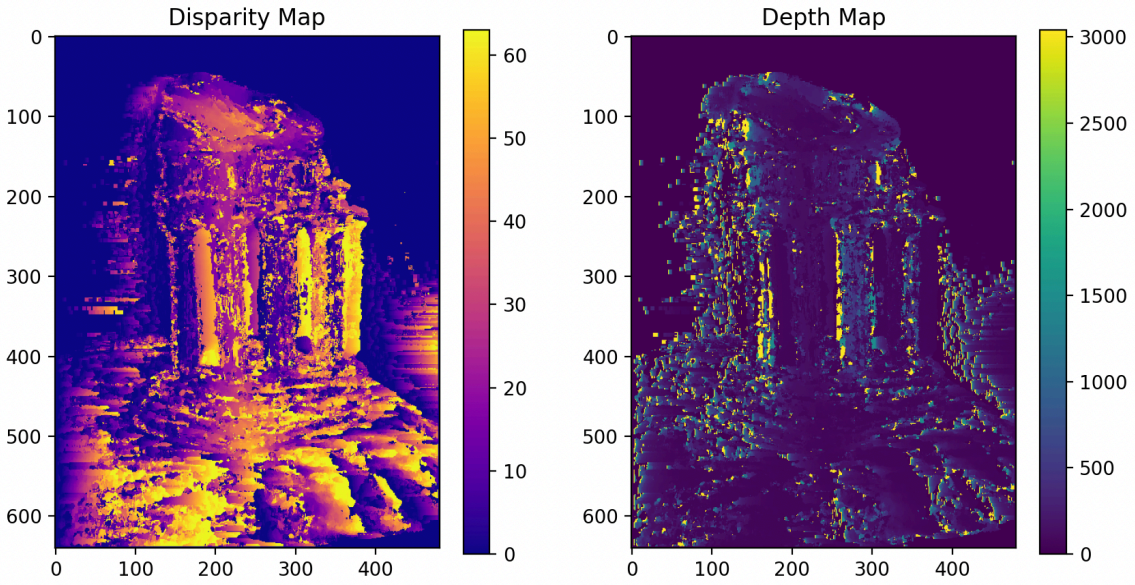
```python
def get_depth(disparity_map, K1, K2, R1, R2, t1, t2):
    baseline = np.linalg.norm(-np.linalg.inv(K1 @ R1) @ (K1 @ t1) - (-np.linalg.inv(K2 @ R2) @ (K2 @ t2)))
    focal_length = K1[0, 0]

    depth_map = np.zeros_like(disparity_map)
    valid_disp = disparity_map > 0
    depth_map[valid_disp] = (baseline * focal_length) / disparity_map[valid_disp]

    return depth_map
```

It computes baseline → Distance between the cameras

Extract the focal length and compute depth
.4.Draw Epipolar Coordinates (draw_epipolar_lines function

```python
def draw_epipolar_lines(image, num_lines=10):
    h, w = image.shape
    step = h // num_lines
    for i in range(0, h, step):
        cv2.line(image, (0, i), (w, i), (255, 0, 0), 1)
    return image

# Load rectified images and draw epipolar lines
rectified_im1 = draw_epipolar_lines(im1.copy())
rectified_im2 = draw_epipolar_lines(im2.copy())

# Display rectified images with epipolar lines
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(rectified_im1, cmap="gray")
plt.title("Rectified Image 1 with Epipolar Lines")

plt.subplot(1, 2, 2)
plt.imshow(rectified_im2, cmap="gray")
plt.title("Rectified Image 2 with Epipolar Lines")

plt.show()
```

FINAL IMAGES ARE:


Disparity Map
0
100
200
300
400
500
600
0
100
200
300
400
60
50
40
30
20
10
0
Depth Map
0
100
200
300
400
500
600
0
100
200
300
400
3000
2500
2000
1500
1000
500
0